

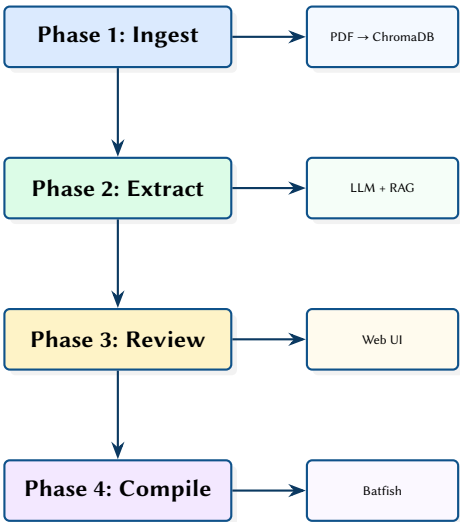
LLM-Powered Network Configuration Parsing and Cross-Vendor Compilation

RAG-Based Extraction, Human-in-the-Loop Review, and Semantic Validation

Simon Knight

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Abstract. This report presents a network automation framework that decouples configuration from vendor-specific syntax using LLM-powered Retrieval-Augmented Generation. The system operates in four phases: vendor documentation ingestion into a ChromaDB vector store, LLM-powered topology extraction with confidence scoring and evidence tracking, human-in-the-loop web-based review with citation enforcement, and cross-vendor configuration compilation with Batfish semantic validation. The vendor-neutral intermediate representation models network relationships (interfaces, VLANs, OSPF, BGP) rather than device syntax, enabling the same topology to compile to Cisco IOS and Arista EOS output with full evidence traceability.

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LLM	Large Language Model	2
RAG	Retrieval-Augmented Generation	2
IR	Intermediate Representation	2
HITL	Human-in-the-Loop	5
OSPF	Open Shortest Path First	4
BGP	Border Gateway Protocol	4

1 Introduction and Motivation

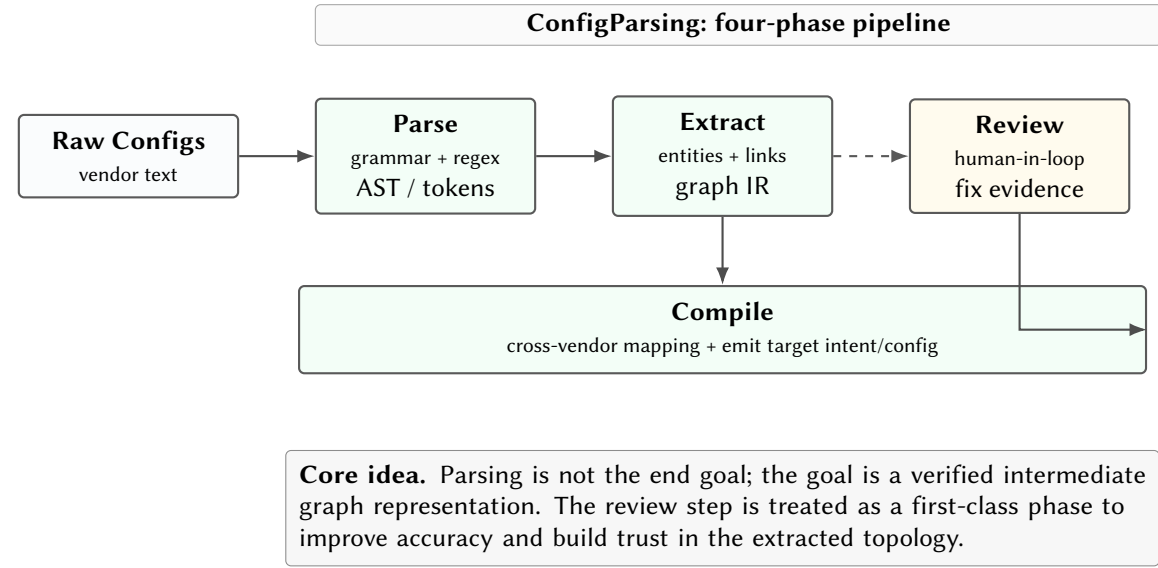


Figure 1. From raw configurations to a validated graph IR and cross-vendor compilation.

Network configuration is inherently vendor-specific: the same logical intent (“interface Ethernet1 has IP 10.0.0.1/30 in VLAN 100”) is expressed differently across Cisco IOS, Arista EOS, Juniper JunOS, and others. Traditional parsers use hand-written regular expressions or TextFSM templates, which are brittle, vendor-locked, and require constant maintenance as CLI syntax evolves.

This framework takes a fundamentally different approach: it uses Large Language Model (LLM)-powered Retrieval-Augmented Generation (RAG) [1] to understand vendor documentation and extract network intent from configurations, producing a vendor-neutral Intermediate Representation (IR) that can be compiled to any target vendor.

: Evidence-Backed Extraction

Every extracted field carries a confidence score (0.0–1.0) and evidence citations linking the value to source configuration lines and documentation chunks. Low-confidence extractions are automatically flagged for human review.

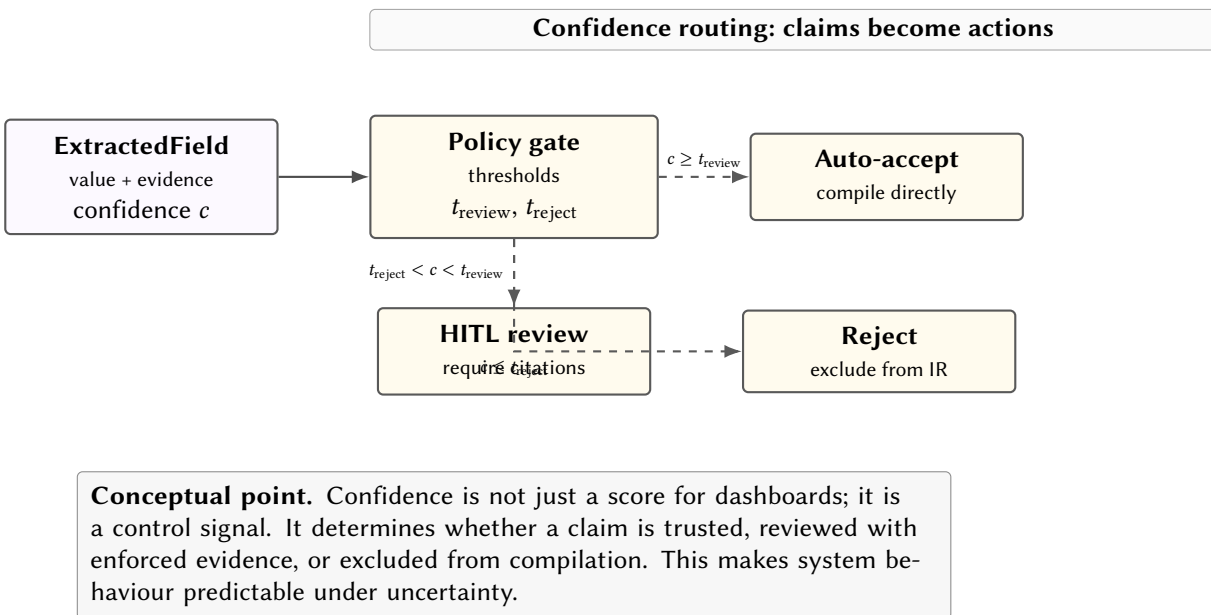


Figure 2. Confidence routing converts extraction uncertainty into operational behaviour: accept, review, or reject with explicit thresholds.

2 Phase 1: Knowledge Base Ingestion

Vendor documentation (PDF manuals, CLI references) is ingested through a layout-aware PDF parser that preserves section structure and code blocks. Parsed chunks are embedded and stored in ChromaDB [2] with deterministic chunk IDs for reproducible citations.

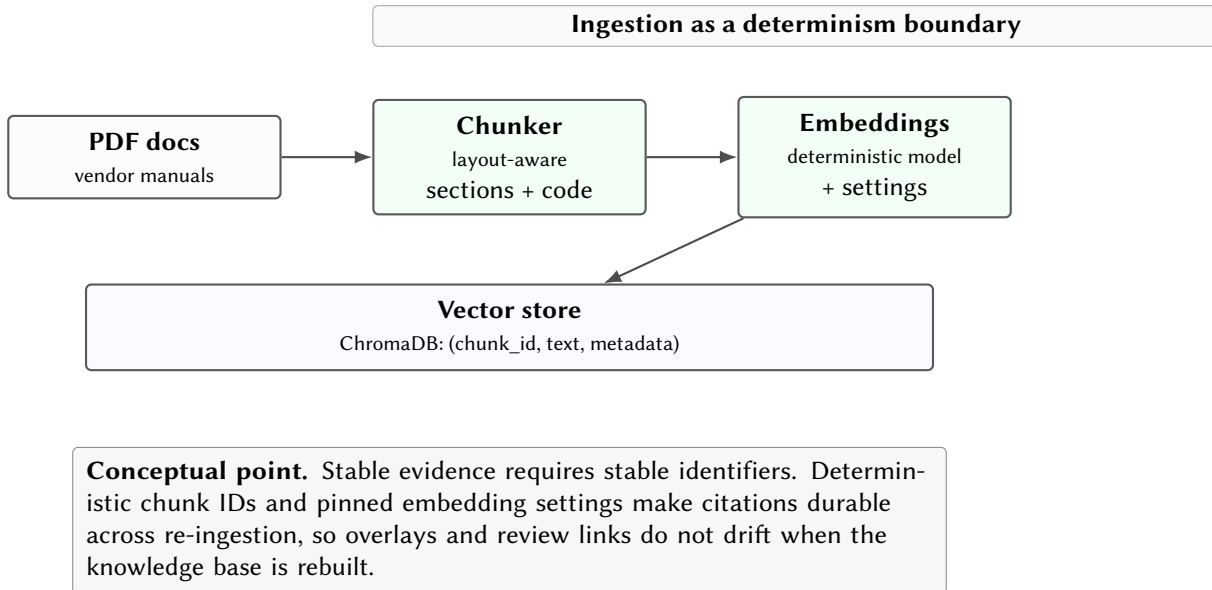
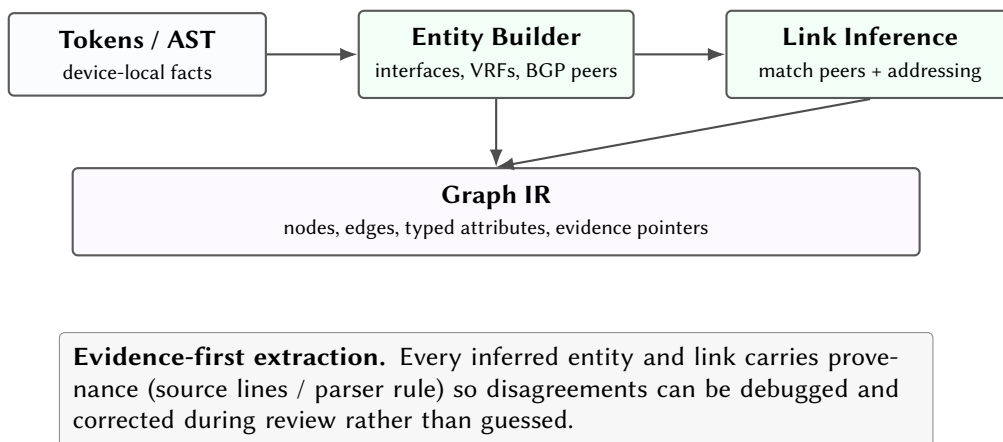


Figure 3. Knowledge-base ingestion as a reproducibility boundary. Deterministic chunk IDs preserve end-to-end evidence chains across re-ingestion.

```
class DocumentIngester:
    def ingest_pdf(self, path: Path, vendor: str) -> int:
        chunks = self.parser.extract_chunks(path)
        for chunk in chunks:
            chunk_id = deterministic_id(vendor, path.name, chunk.page, chunk.index)
            self.collection.add(
                ids=[chunk_id],
                documents=[chunk.text],
                metadatas=[{"vendor": vendor, "page": chunk.page}],
            )
        return len(chunks)
```

3 Phase 2: Topology Extraction



Insight:
Deterministic chunk IDs ensure that re-ingesting the same document produces identical identifiers, preserving evidence citations across extractions without invalidating the overlay database.

Figure 4. Topology extraction as a graph construction problem: facts are lifted into entities, then joined into links with explicit provenance.

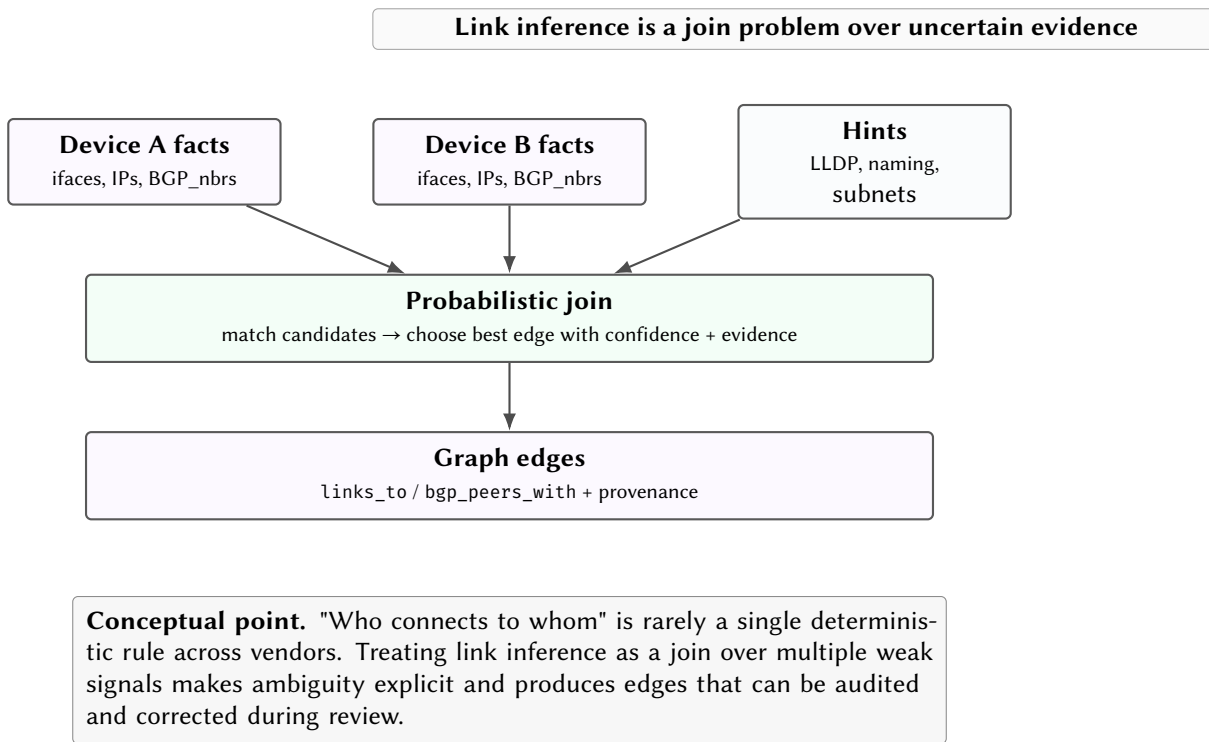


Figure 5. Link inference as a constrained join. Multiple evidence sources contribute to candidate matches; the chosen edge carries confidence and provenance.

The extraction engine processes vendor configurations through an LLM with RAG context from the knowledge base. For each configuration section, relevant documentation chunks are retrieved from ChromaDB and included in the prompt, giving the LLM vendor-specific context for accurate parsing.

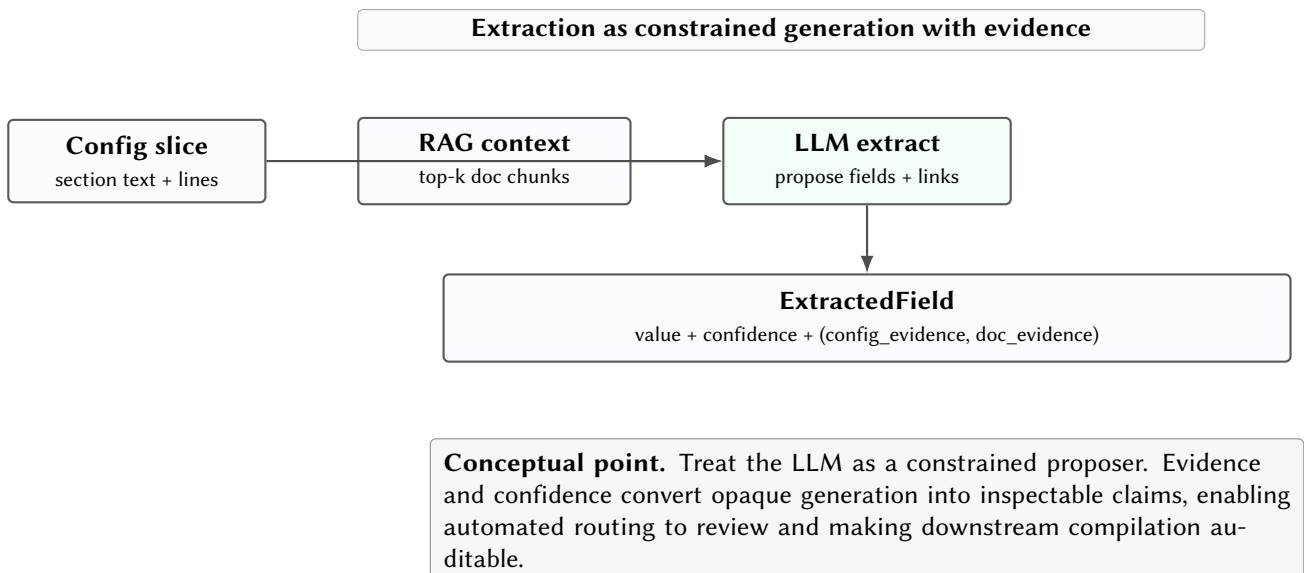


Figure 6. Evidence-backed extraction. Config text and retrieved docs jointly condition the LLM, which outputs values plus provenance, not just a best-guess parse.

The output is a vendor-neutral IR modelling:

- **Interfaces** — name, IP addresses, VLANs, status
- **Routing** — Open Shortest Path First (OSPF) areas, Border Gateway Protocol (BGP) peers, static routes
- **Services** — DHCP, NTP, DNS, ACLs

Each extracted field includes:

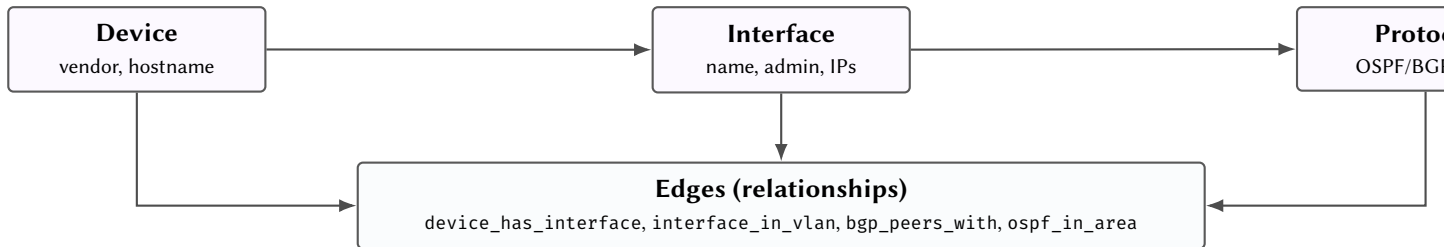
```
@dataclass
class ExtractedField:
```

```

value: Any
confidence: float # 0.0 - 1.0
config_evidence: list[str] # Source config line numbers
doc_evidence: list[str] # ChromaDB chunk IDs

```

The IR is a typed graph schema (not a pile of strings)



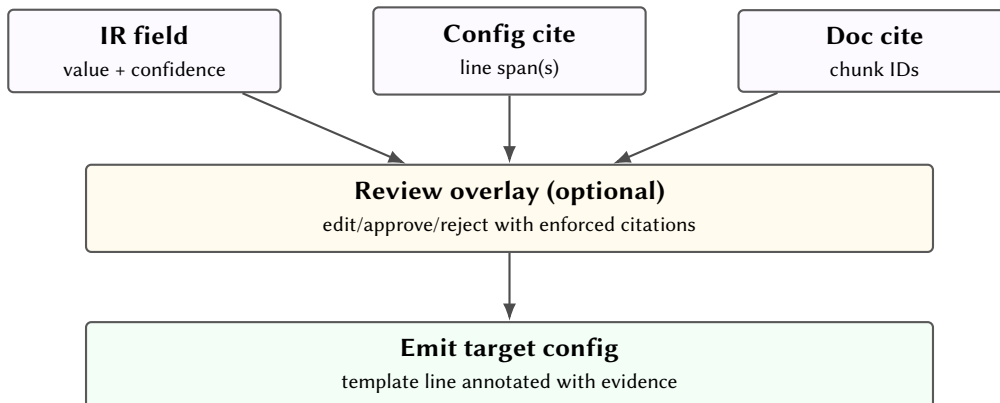
Conceptual point. A typed graph encodes intent as relationships. Compilation and validation operate on this structure (joins, constraints, reachability), which is why the system generalises across vendors better than regex-only parsing.

Figure 7. IR as a schema: entities plus typed relationships. Evidence attaches to nodes/edges, enabling traceable compilation and semantic checks.

Design Decision 1: Hallucination Protection

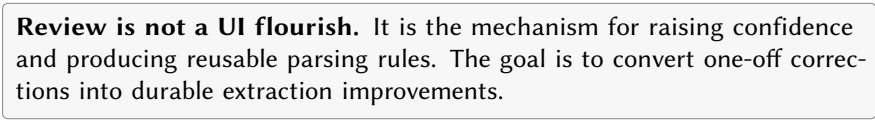
Fields extracted with confidence below the configurable threshold (default 0.6) are automatically flagged for Human-in-the-Loop (HITL) review. The evidence chain (config line → doc chunk → extracted value) enables reviewers to verify the LLM's reasoning rather than trusting the output blindly.

Evidence chain anatomy: from claim to emitted config



Conceptual point. Evidence is a first-class payload. It must survive extraction, review, and compilation so any emitted line can be traced back to concrete sources. This prevents "silent drift" where a config compiles but no one can justify why.

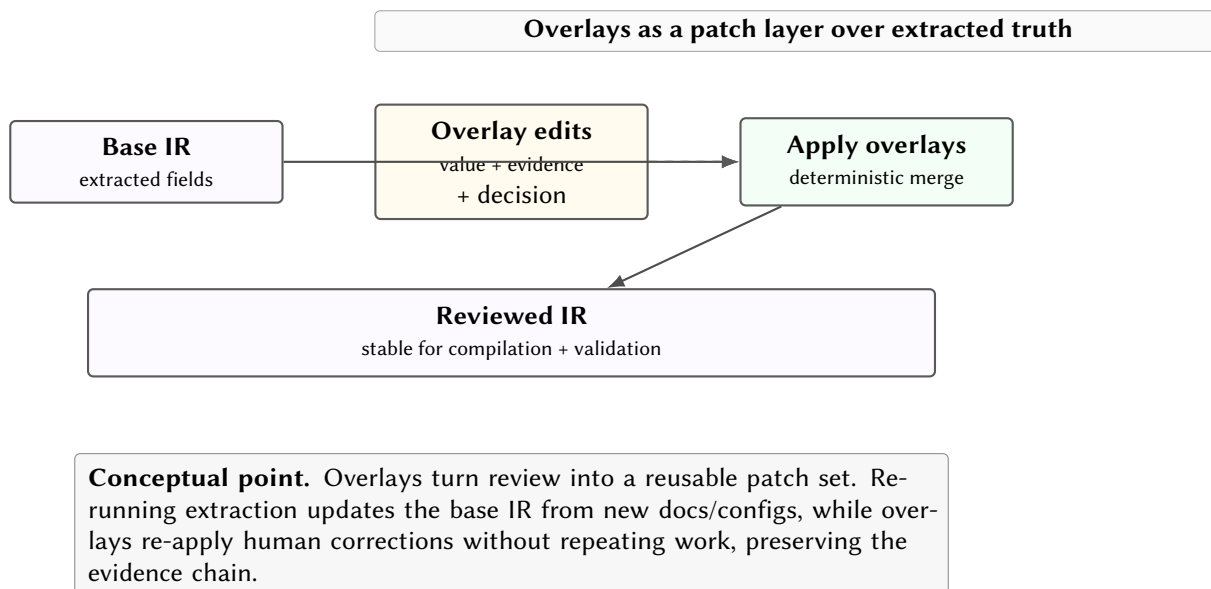
Figure 8. End-to-end evidence chain. The same provenance objects support auditing, review enforcement, and annotated compilation outputs.



A web-based review interface presents flagged extractions with their evidence chains. Reviewers can:

-
- ```
graph TD; Title[RAG prompt composition and failure modes]; Config[Config slice
bounded section]; Retrieved[Retrieved docs
top-k + rerank]; Constraints[Constraints
schema + "cite" rules]; Prompt[Prompt
slice + context + constraints within token budget]; Failure[Failure modes
truncation, wrong retrieval, stale docs, ambiguous CLI]; Config --> Prompt; Retrieved --> Prompt; Constraints -.-> Prompt; Prompt -.-> Failure;
```
- RAG prompt composition and failure modes**
- Config slice**  
bounded section
- Retrieved docs**  
top-k + rerank
- Constraints**  
schema + "cite" rules
- Prompt**  
slice + context + constraints within token budget
- Failure modes**  
truncation, wrong retrieval, stale docs, ambiguous CLI
- Conceptual point.** Many extraction errors are retrieval or budget errors, not "LLM errors". Making prompt composition explicit helps debug issues: fix chunking/reranking, reduce slice size, or tighten constraints before changing the model.

Approved edits are stored as overlays that persist across re-extractions. When the extraction pipeline runs again (e.g. after documentation updates), approved overlays are applied automatically, preserving human corrections.



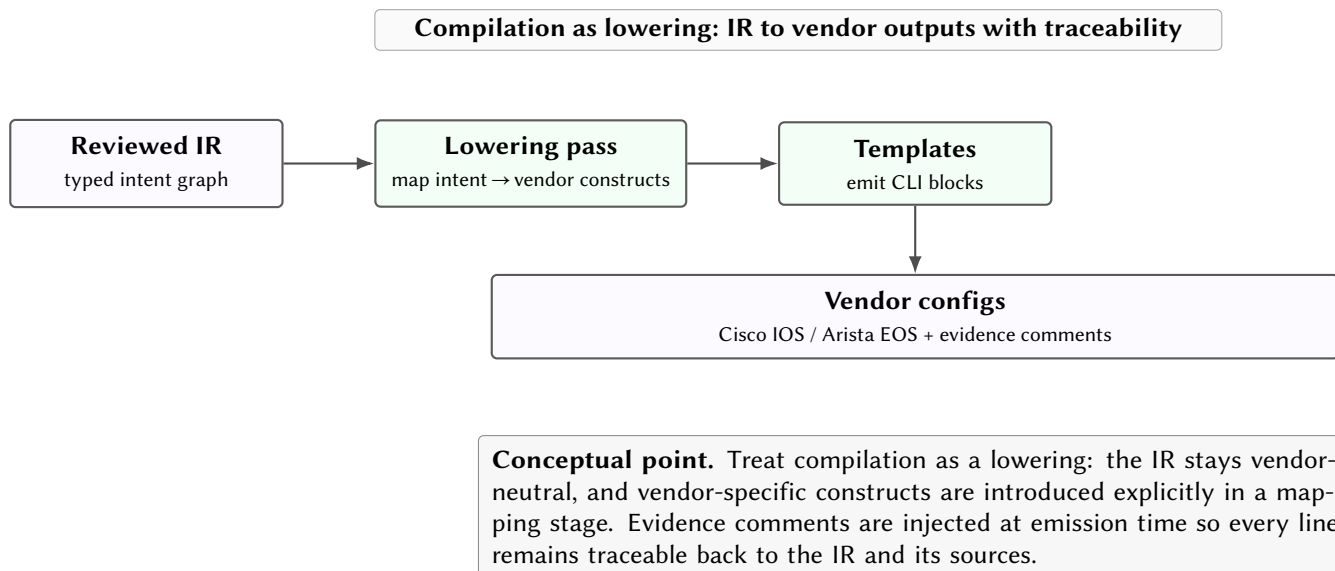
**Figure 11.** HITL review produces overlays: durable patches applied deterministically on top of regenerated extraction output.

#### : Evidence Enforcement

All manual edits must cite either a configuration line number or a documentation chunk ID. This prevents unsupported assertions from entering the IR and maintains the end-to-end evidence chain.

## 5 Phase 4: Cross-Vendor Compilation

The compiler transforms the vendor-neutral IR into target vendor configurations using Jinja2 templates. Currently supported targets are Cisco IOS and Arista EOS.



**Figure 12.** Cross-vendor compilation as a lowering pipeline. Vendor constructs are introduced explicitly (mapping), then rendered (templates) with evidence annotations.

Generated configurations include evidence comments tracing each line back to the IR field and its original source:

```

! Source: interface Ethernet1 (confidence: 0.95)
! Evidence: config line 42, doc chunk cisco-ios-ref:p234:3
interface Ethernet1
 ip address 10.0.0.1 255.255.255.252
 no shutdown

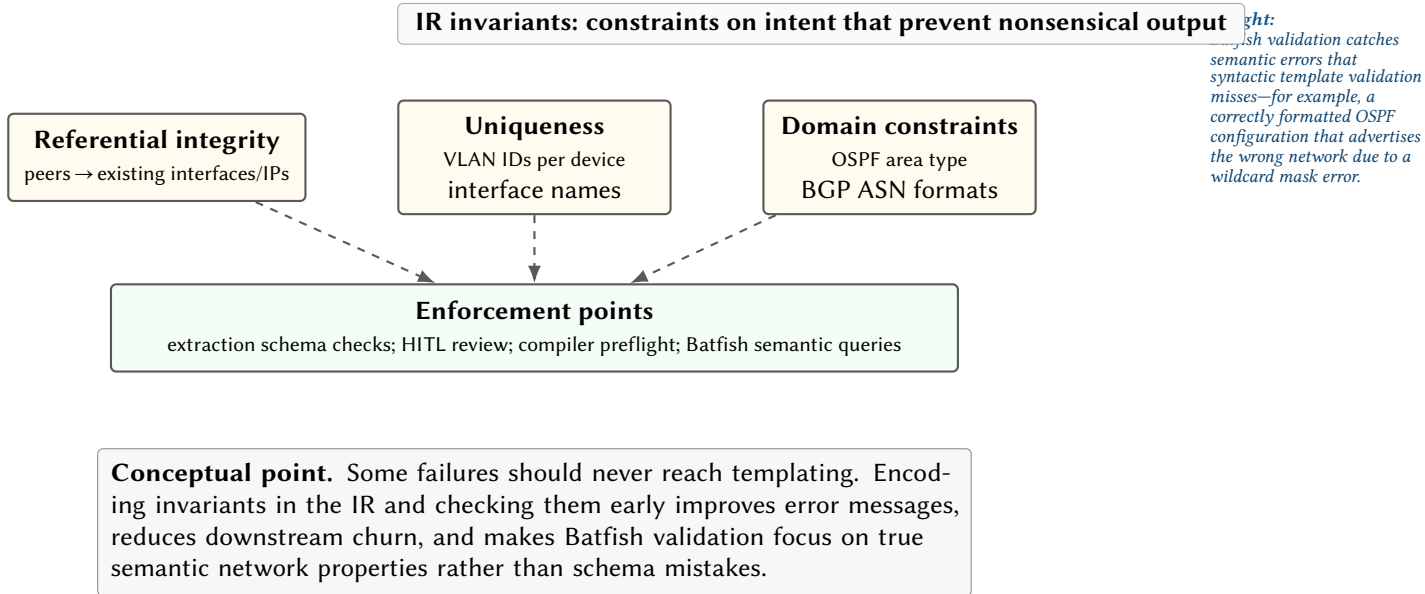
```



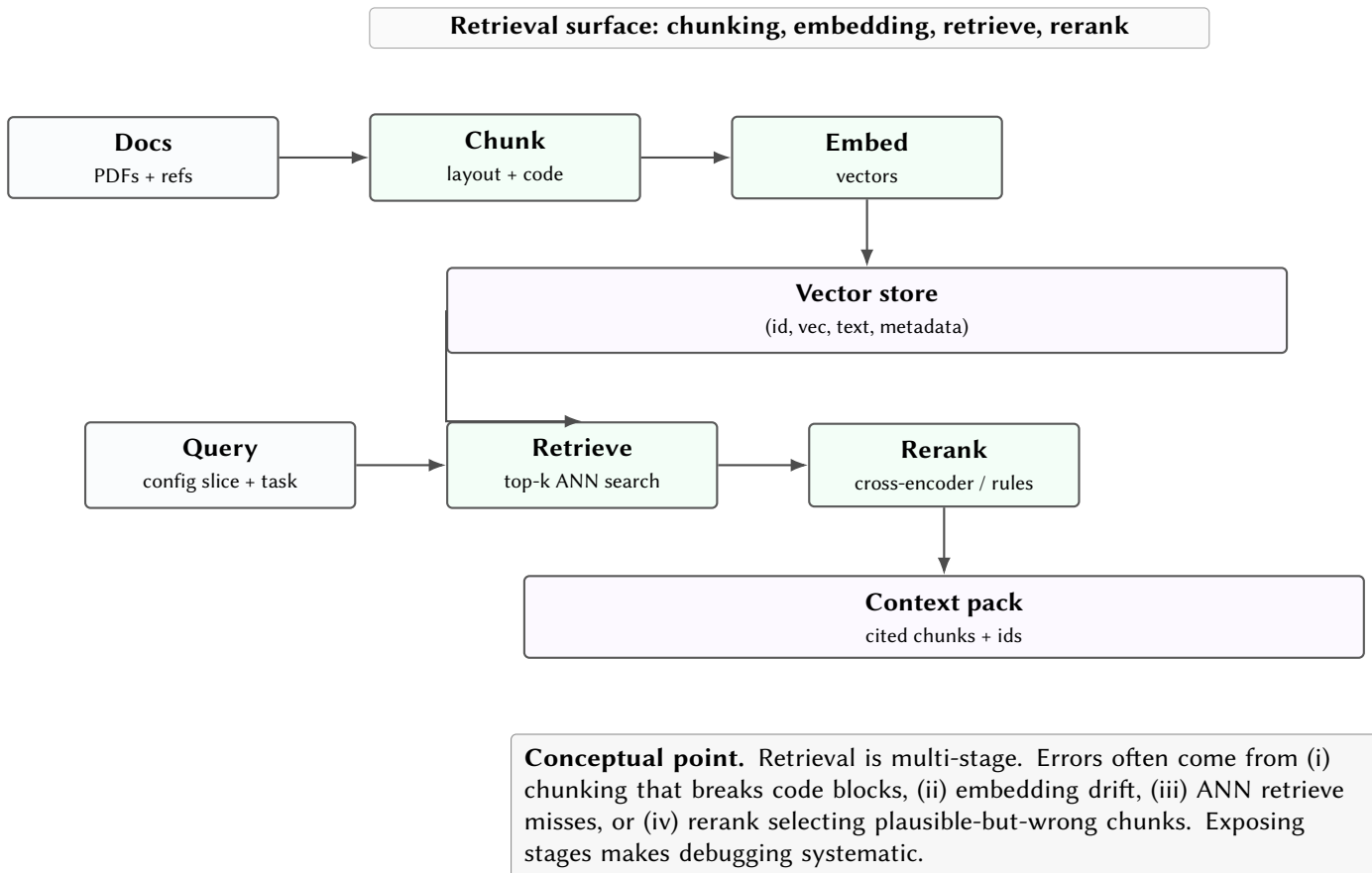
## 5.1 Semantic Validation

Compiled configurations are validated using Batfish [3], which performs semantic analysis to verify:

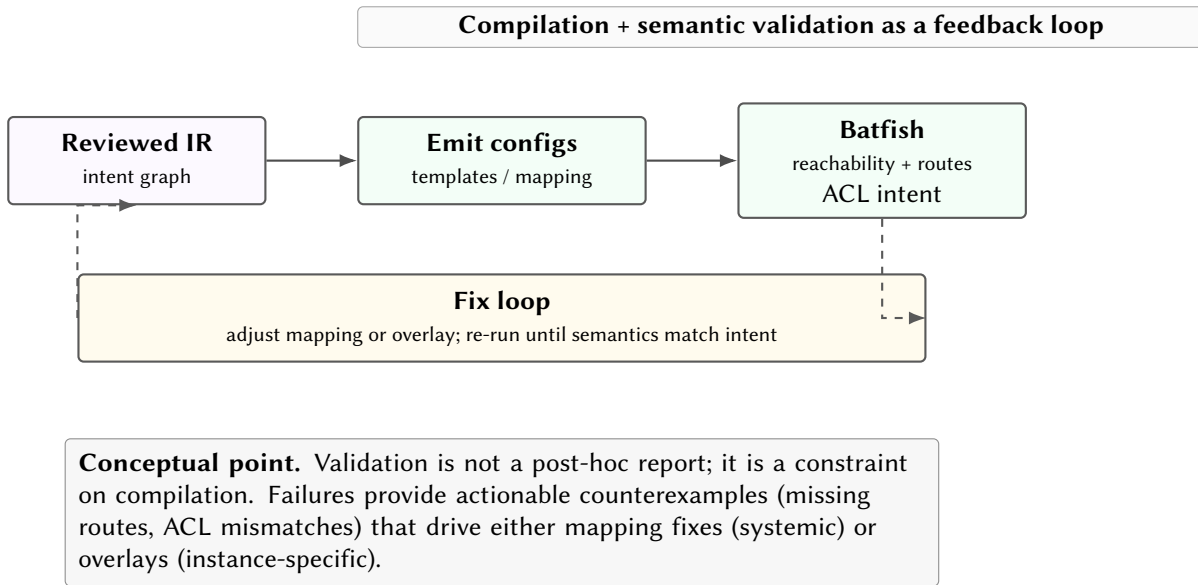
- Reachability between expected endpoint pairs
- Route table correctness (expected prefixes present)
- ACL behaviour matches intent
- No unintended route leaking between VRFs



**Figure 13.** IR constraints and where they are enforced. Invariants provide early failure and clearer debugging than post-hoc syntax checks.



**Figure 14.** RAG retrieval surface. Chunking and reranking are first-class levers for accuracy and explainability.

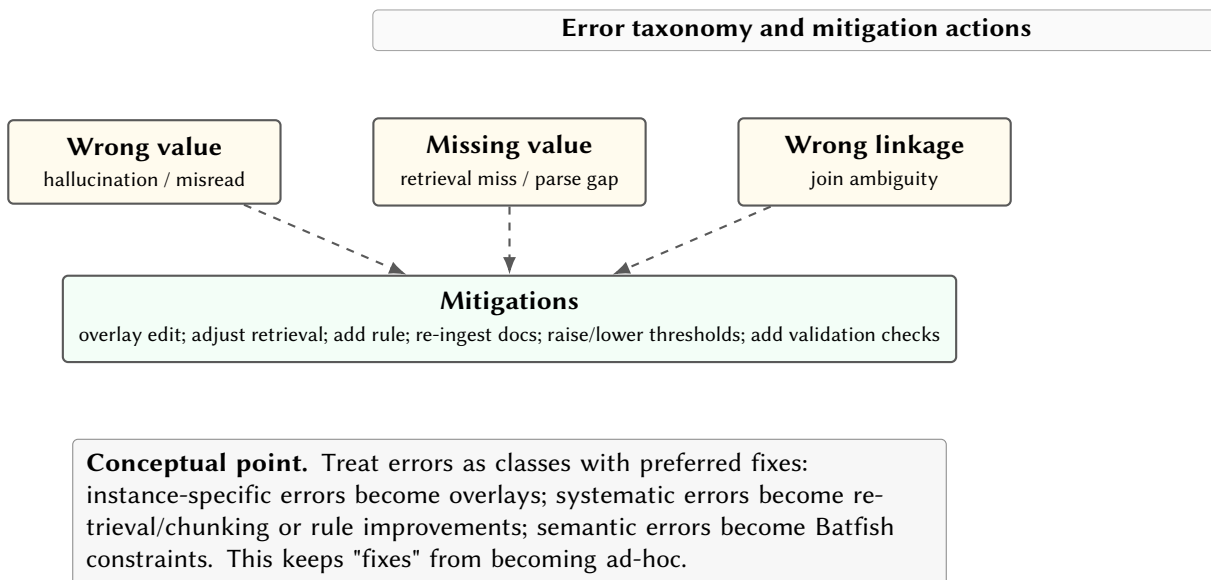


**Figure 15.** Semantic validation closes the loop. Batfish turns emitted configs into semantic facts, which must satisfy the IR’s intent.

## 6 Design Summary

**Table 1.** Key design decisions.

| Decision                | Rationale                                                          |
|-------------------------|--------------------------------------------------------------------|
| RAG over fine-tuning    | Vendor docs change frequently; RAG adapts without retraining       |
| Confidence scoring      | Quantifies extraction reliability; enables automatic HITL routing  |
| Deterministic chunk IDs | Stable citations across re-ingestion; overlays survive doc updates |
| Overlay persistence     | Human corrections survive re-extraction without manual re-review   |
| Batfish validation      | Semantic verification catches errors invisible to syntactic checks |



**Figure 16.** Operational taxonomy: common failure modes and the most effective mitigation levers in the pipeline.

## References

- [1] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive nlp tasks,” *NeurIPS*, 2020.
- [2] Chroma, *ChromaDB: The open-source embedding database*, 2024. [Online]. Available: <https://www.trychroma.com>

- [3] Intentionet, *Batfish: Network configuration analysis*, 2024. [Online]. Available: <https://www.batfish.org>

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Revision History

| Version | Date       | Changes                                   |
|---------|------------|-------------------------------------------|
| 1.0.0   | March 2026 | Initial report (all four phases complete) |